EDR Explained

The eddy dissipation rate: what it is, how aviation uses it, and how MODE-S Wind estimates it from Mode S data

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In short. Turbulence is the bumpy, irregular part of the wind. **EDR**, the *eddy dissipation rate*, is one number that says how strong it is — a property of the *air*, so the same value describes the air for every aircraft that flies through it. It is the turbulence measure of ICAO: an EDR below 0.10 is smooth air, from **0.10 light**, from **0.20 moderate** and from **0.45 severe** turbulence (for a medium-size airliner). MODE-S Wind estimates an EDR for every landing and departure at Helsinki-Vantaa from the speeds the aircraft report in their Mode S replies about every two seconds, and shows it on the Windshear page, in the History lists and on the approach map of the phone view. This document explains the idea in plain words (Section 1), the physics behind it (Section 2), its history and use in aviation (Sections 3–5), how MODE-S Wind computes it (Section 6) and what it has found so far (Section 7).

A research estimate. The MODE-S Wind EDR is computed by a passive receiver from the aircraft's routine replies, over scales of 100–200 m and averaged over a whole approach or climb. It is not the one-minute peak EDR that aircraft systems report under ICAO rules, it is not certified, and it must not be used for flight operations or air traffic control. It is a tool to study the turbulence of the lowest kilometre at the airport, and it will be calibrated against pilot reports as they are collected. This document is updated whenever the calculation changes (Section 10).

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1 EDR in plain words

Turbulence. The wind is never perfectly steady. Besides its mean speed and direction it contains *eddies* — swirls of air of every size, from kilometres down to millimetres. An aircraft flying through them is pushed up and down and sideways: the passengers feel bumps, the airspeed jumps by a few knots, the autopilot works harder. Near the ground the eddies come mostly from the wind blowing over buildings, forests and hills, and from the sun warming the ground; higher up they come from wind shear, waves over mountains, jet streams and convective clouds.

Why one number is hard. How strongly an aircraft is shaken depends not only on the air but also on the aircraft: a small, light aircraft at low speed is thrown about much more than a heavy jet in the same air. For decades pilots reported turbulence as "light", "moderate" or "severe" from what they felt, and these words meant different things on different types — a report from a turboprop could not be compared with one from a wide-body.

The idea of EDR. The large eddies get their energy from the wind, break up into smaller ones, which break up into still smaller ones, until the smallest turn the motion into heat. In the middle of this chain — the *cascade* — the rate at which energy flows down to heat is the same at every eddy size. That rate, the *dissipation rate* ε, is a property of the air alone. EDR is its cube root, $\varepsilon^{1/3}$, which grows roughly in proportion to how strong the gusts are. Any aircraft, a lidar or a radar can measure it, and the result describes the air, not the vehicle.

The scale. EDR is given in $\mathrm{m}^{2/3}\,\mathrm{s}^{-1}$ and ranges from 0 (perfectly smooth air) to about 1 (extreme turbulence). ICAO uses these limits for a medium-size transport aircraft:

EDR	ICAO turbulence	What it feels like (medium airliner)
below 0.10	**smooth** (none)	no or barely noticeable bumps
0.10–0.20	**light**	slight bumps, drinks ripple; seat belts optional
0.20–0.45	**moderate**	clear jolts, walking is difficult, loose objects move
0.45 and more	**severe**	large, abrupt changes; momentary loss of control possible

The descriptions are the familiar pilot-report wording, given only as an illustration; the official limits refer to the *peak* EDR of the reporting interval (Section 4).

In MODE-S Wind. The receiver near Tuusula hears every aircraft landing at or departing from Helsinki-Vantaa answering the ground stations with its true airspeed and ground speed about every two seconds. Their difference is the headwind the aircraft is flying into. When the air is turbulent, this headwind jumps up and down from one reply to the next; when the air is smooth, it changes only slowly. MODE-S Wind measures how strongly it jumps over 100–200 m of path, takes away the jumps caused only by the rounding of the speeds in the replies, and turns the rest into an EDR with the physics of the cascade. Every landing and departure gets one value, one below and one above 1 000 ft, and a colour on the map for each straight 2-km piece of its path, in the ICAO colours: green, yellow, orange, red.

2 The physics

2.1 The energy cascade and the inertial subrange

Kolmogorov's theory of 1941 [1] describes turbulence at high Reynolds number as a cascade: energy is fed in at the large scales L (in the atmospheric boundary layer hundreds of metres to kilometres), transferred without loss through ever smaller eddies, and dissipated by viscosity at the Kolmogorov scale η (millimetres). Between the two, in the *inertial subrange* $\eta \ll r \ll L$, the statistics of the velocity differences depend only on the dissipation rate ε ($\mathrm{m}^2\,\mathrm{s}^{-3}$) and the separation r. Dimensional analysis then gives the energy spectrum and the second-order structure function of the velocity component u along the separation (Fig. 1):

$$E(k) = C_K\,\varepsilon^{2/3}k^{-5/3}, \qquad D_{uu}(r) = \left\langle [u(x+r)-u(x)]^2 \right\rangle = C\,\varepsilon^{2/3}r^{2/3}, \tag{1}$$

with the Kolmogorov constant $C_K \approx 0.5$ for the one-dimensional longitudinal spectrum and $C \approx 2.0$–2.1 for the longitudinal structure function [3]. In practice the inertial law is fitted together with an outer scale, usually in the form of the von Kármán spectrum [2, 4]. The eddy dissipation rate used in aviation is

$$\mathrm{EDR} = \varepsilon^{1/3} \qquad (\mathrm{m}^{2/3}\,\mathrm{s}^{-1}), \tag{2}$$

which scales linearly with the velocity fluctuations: from Eq. (1), the root-mean-square velocity difference over a distance r is $\sqrt{C}\,\mathrm{EDR}\,r^{1/3}$ — with EDR = 0.2 about 1.3 m/s (2.5 kt) over 100 m and 2.8 m/s (5.5 kt) over 1 km.

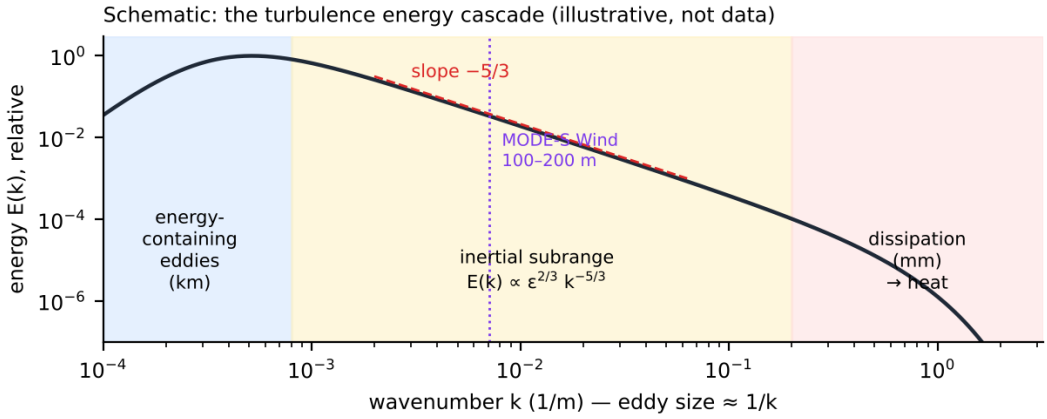


Figure 1. The energy cascade (schematic, not measured data). Energy enters at the scale of the large eddies, flows through the inertial subrange with the $-5/3$ slope of Kolmogorov, and is dissipated at millimetre scales. EDR measures the level of the $-5/3$ part. The dotted line marks the 100–200 m scale that the Mode S replies resolve on final approach (Section 6).

2.2 Why EDR is a property of the air

An aircraft responds to the turbulence through its own dynamics: its vertical acceleration for a given gust depends on its wing loading, speed and altitude, and on how its response function weights the gust spectrum. The *intensity* of the air turbulence is however fully described by ε in the inertial subrange; given ε, the response of a particular aircraft can be computed. This is why EDR was chosen as the aircraft-independent metric (Section 3), and why the ICAO limits are tied to an aircraft size category: the same EDR is "light" for a heavy aircraft and "moderate" for a light one [7].

2.3 Mean and peak, and the averaging scale

EDR estimated over a long record is a mean; turbulence is intermittent, and the strongest moments within it are much rougher. Aircraft algorithms therefore report the *median* and the *peak* of the EDR estimates within each reporting interval (one minute) [7]. The MODE-S Wind value is an average over a whole approach or climb, or over a 2-km piece of it, and is thus closer to a mean than to a one-minute peak — a reason it reaches the ICAO limits less often than a peak would (Section 7).

3 A short history: from "bumps" to EDR

Pilot reports and gust loads. For most of aviation history turbulence was reported by pilots in words — light, moderate, severe — from what they felt. These reports are subjective, sporadic and depend on the aircraft type [6, 30]. Engineers measured gusts through their effect on the aircraft: the *derived equivalent vertical gust* (DEVG), still reported in the AMDAR programme mainly from the Southern Hemisphere, is in effect a gust-load transfer factor of the aircraft rather than a direct measure of the air [11].

MacCready (1964). MacCready proposed to standardise "gustiness" by a property of the atmosphere rather than of the aircraft: one intensity number, measurable continuously in flight, to which the root-mean-square vertical acceleration of a given aircraft at a given speed is related linearly [5]. Founded on the inertial subrange, this is the idea behind EDR.

NCAR and the airlines (1995–2014). Cornman, Morse and Cunning showed how the turbulence intensity can be estimated in real time from the routine data of an airliner within the limits of its avionics [6]. Two NCAR algorithms followed [7]: one from the vertical acceleration with a model of the aircraft's response (United B737 and B757 from 2001, reporting the median and 90th percentile of each minute), and one from the vertical wind computed by the aircraft, which needs no response model (Delta B737 from 2008, reporting the mean and peak of each minute). Both estimate EDR over nominal one-minute intervals from overlapping 10-second windows. By 2014 about 200 aircraft had produced more than 137 million EDR reports [7]; about 1 600 aircraft of several airlines report in-situ EDR today [15]. Comparing the reports with pilot reports, Sharman et al. found median EDRs of about 0.22 for moderate and 0.47 for severe turbulence on medium-size aircraft, and that EDR is lognormally distributed at all altitudes [7].

An ICAO standard (2001). In 2001 EDR became the ICAO standard metric for automated turbulence reports from aircraft [9, 11]. Its limits were revised by Amendment 79 to Annex 3 (applicable 5 November 2020), following the comparison with pilot reports (Section 4).

4 EDR in aviation today

4.1 Reports from aircraft

ICAO reports turbulence as the cube root of the eddy dissipation rate. Since Amendment 79 the categories, for the *peak* EDR of a medium-size transport aircraft in typical en-route conditions, are [12, 13]:

Category	Peak EDR ($m^{2/3}\,s^{-1}$)	Before Amendment 79 (until 2020)
nil	≤ 0.10	≤ 0.10
light	0.10–0.20	0.10–0.40
moderate	0.20–0.45	0.40–0.70
severe	≥ 0.45	≥ 0.70

A special air-report is due when the peak EDR reaches 0.20 (before: 0.40), and the SIGMET criteria use the same moderate and severe limits [12]. A note to the provisions recalls that the turbulence actually felt depends on the aircraft type, mass, altitude, configuration and airspeed. The detailed provisions moved in 2025 to the *Procedures for Air Navigation Services – Meteorology* (PANS-MET, Doc 10157, applicable 27 November 2025) with the limits unchanged: routine en-route reports give the mean and the peak of the preceding 15 minutes and the minute of the peak, special air-reports the mean and peak of one minute, and during the first 10 minutes of the climb the peak over 30 seconds [13]. Most implementations agree well on the one-minute mean; the peak depends more on the window length, and the noise floor of the aircraft algorithms is about 0.01–0.02 [10].

4.2 Forecasts

EDR is also the unit of the turbulence forecasts. The Graphical Turbulence Guidance (GTG) of NCAR maps each turbulence diagnostic of a weather model to EDR, assuming a lognormal distribution fitted to the climatology of the in-situ peak EDRs, and combines them into an ensemble mean that is verified against in-situ EDR and pilot reports [8]. Since November 2020 the World Area Forecast System (WAFS) issues gridded turbulence forecasts in EDR (0–1) on 0.25° grids from FL100 to FL450, based on GTG and harmonised between WAFC London and Washington [14].

4.3 The same air, different aircraft

EDR describes the air; the response of an aircraft depends on its size, weight, speed, altitude and aerodynamics, through an aircraft-specific response function that relates the rms vertical acceleration linearly to EDR [7]. ICAO gives limits only for medium-size aircraft: a heavy aircraft needs a higher EDR, a light one a lower EDR to feel the same, and type-specific values are left to the aircraft documentation [14].

4.4 Near the ground

Most in-situ EDR reports come from cruise. Their vertical profile has maxima near the surface and around 11 km and a minimum in the mid-troposphere, and the share of peak EDRs above 0.2 is higher at low levels [7]; in all only about 10 % of the reports exceed 0.1. In the boundary layer, lidar climatologies find daytime convective turbulence of EDR about 0.2 in summer and 0.08 in winter, and turbulence driven by the shear of low-level jets of about 0.1 [18]. Under a nocturnal jet the turbulence is produced by the shear below the jet nose and is weakest in its core, and in the weakly stable boundary layer its strength depends mainly on the jet [19]; very low EDRs go with strong stability [20].

5 EDR at Hong Kong International Airport

The Hong Kong Observatory (HKO) has the longest operational experience with EDR in the terminal area. Turbulence at HKIA occurs mainly when the wind blows over the hills of Lantau Island south of the airport [24]. The Windshear and Turbulence Warning Service uses a terminal Doppler weather radar, four long-range and eight short-range Doppler lidars, automatic weather stations on hilltops and in valleys, buoys and wind profilers, and issues alerts for the area within 3 NM of the runway ends, updated at least once a minute [21].

- **Anemometers and radar.** The Windshear and Turbulence Warning System (WTWS) estimated EDR along the glide paths from a regression on the anemometer winds (weight 0.8) and a radar-based algorithm (0.2), with an alert threshold of EDR 0.3 and, in the HKO papers of 2004–2008, moderate turbulence at 0.3–0.5 and severe from 0.5 — local values that differ from both the old and the present ICAO limits [22, 23]. The anemometer regressions were re-derived from the flight data (quick access recorder, QAR) of 832 arrivals, processed to 4-Hz EDR with the 90th percentile per approach [27].
- **Lidar.** Chan derived EDR maps from the plan-position scans of the airport lidar: in sectors of about 1 km by 20° a plane is fitted to the radial velocities and the structure function of the residuals is fitted to the von Kármán model; the maps were checked against a hilltop sonic anemometer [24]. Glide-path lidar scans were compared with the WTWS

and pilot reports [23], the lidar outperformed the anemometer method for moderate-or-greater turbulence in typhoon Nesat (QAR of 423 jets, [25]), and lidar-derived EDR profiles are used in the low-level alerts [26].

- **Aircraft data.** HKO collects automatic EDR downlinks, QAR data and turbulence special air-reports (112 in 2025) [30]. From ADS-B it derives the rms vertical acceleration (RMSVA) of each aircraft: at low level over 5-second windows on final approach, with a weak but significant correlation with lidar EDR on runways 07L/07R [28], and en route as an operational intranet product since November 2024 (62 % detection against 122 special air-reports) [29].

Two lessons carry over to Helsinki-Vantaa: EDR from different sensors and scales agrees only after calibration against a common reference — at Hong Kong flight data and pilot reports — and local alert thresholds are chosen for the local climate and operations.

6 EDR from Mode S replies: the MODE-S Wind method

6.1 The measurement

Mode S Enhanced Surveillance ground stations interrogate each aircraft for registers filled from its flight computers. Register **BDS 5,0** (track and turn report) contains, from the same instant, the roll angle, true track, ground speed (GS) and true airspeed (TAS). The difference

$$h = \mathrm{TAS} - \mathrm{GS} \;=\; W_x + \mathrm{TAS}\,(1 - \cos\beta) \tag{3}$$

is the headwind component W_x along the track, plus a small, slowly varying crab-angle term. On final approach to Helsinki-Vantaa an aircraft is asked for BDS 5,0 about every two seconds by the wide-area multilateration and the radars, a new value arriving about every 140–290 m of path. Both speeds are rounded to 2 kt in the register, so a single value of h has a quantisation error of about 1.2 kt (0.6 m/s). No heading, magnetic declination or wind-vector calculation enters, and therefore none of their errors.

MODE-S Wind keeps every BDS 5,0 and 6,0 reply of each landing (and since 27 October 2026 of each departure) in a raw series with the landing record, so the calculation can be repeated whenever it is improved.

6.2 Earlier work with surveillance data

Kopeć et al. [16] tested three ways of retrieving turbulence from Mode S and ADS-B on a research aircraft (Cessna Citation II, 2013): the vertical acceleration from ADS-B altitude or vertical rate, the structure function of Mode S EHS winds ($C = 2$) and a threshold-crossing method. The Mode S wind methods separated light from moderate turbulence without false alarms but underestimated the magnitude, at a radar rate of one reply per 4 seconds. Olive and Sun [17] used the difference between the barometric and inertial vertical rates of BDS 6,0 as a turbulence indicator (a proof of concept without EDR); NCAR inverts the ADS-B vertical rate (1 Hz, 64-ft/min resolution) into EDR without aircraft response models [15]; and HKO uses the ADS-B vertical rate as described above. MODE-S Wind differs in its data: at Helsinki-Vantaa the wide-area multilateration asks the aircraft on final for BDS 5,0 about every two seconds, so the headwind is sampled every 140–290 m in the lowest 3 000 ft — the structure function approach of Kopeć et al. applied at twice their rate and in the boundary layer.

6.3 From the headwind series to EDR

The method follows the structure-function approach of Kopeć et al. [16] for Mode S data, adapted to the low-level, high-rate data of an approach:

1. **Series.** The headwind h of every BDS 5,0 reply between 200 and 3 000 ft above the runway threshold (departures: up to 5 000 ft) and the along-track distance x from the ground speed.
2. **Clean times** (method v2, from 31 October 2026). A reply whose time repeats an earlier one is dropped, and so is every copy of a time that lies more than 3 s after a reply stored later. The receiver had stamped some replies up to 50 s ahead of the clock, and the climb-out series then took such a reply again at every sweep, each time at the new height: one departure (FIN3A, 30 October) read 0.34 instead of 0.07. Since 31 October the receiver also sets every message time to the clock when it is ahead, so new series no longer contain such rows.
3. **Straight pieces.** The series is cut into pieces of about 2 km; only pieces in which the track changes by no more than 15° are used. In a turn the headwind part of the *mean* wind changes by itself and would look like turbulence (the first version without this gate correlated with the turns rather than with the weather).
4. **Detrending.** Each piece is detrended with a straight line, so that the slow change of the mean wind with height does not count.
5. **Structure function.** For all pairs of replies of a piece separated by r = 100–200 m, $D = \langle(\Delta h)^2\rangle$.
6. **Noise floor.** The 2-kt rounding of TAS and GS alone gives $D = N_0 = 0.35\ \mathrm{m^2\,s^{-2}}$; this is also the value of the calmest days. It is subtracted.

7. **Inertial scaling.** With Eq. (1) at the middle of the separation bin, $r = 141$ m and $C = 2$:

$$\mathrm{EDR} = \sqrt{\frac{D - N_0}{2 \cdot 141^{2/3}}}\,. \tag{4}$$

Per flight the pairs of all straight pieces are pooled into one value (*EDR*), and separately those of the pieces below and above 1 000 ft above the threshold; the strongest piece is kept as the *peak*, and every piece with its EDR, height and position for the map. A flight needs at least 8 replies in a piece, 3 pairs per piece and 6 pairs in all; otherwise it has no EDR ("no EDR" in the lists). Fig. 2 shows two landings.

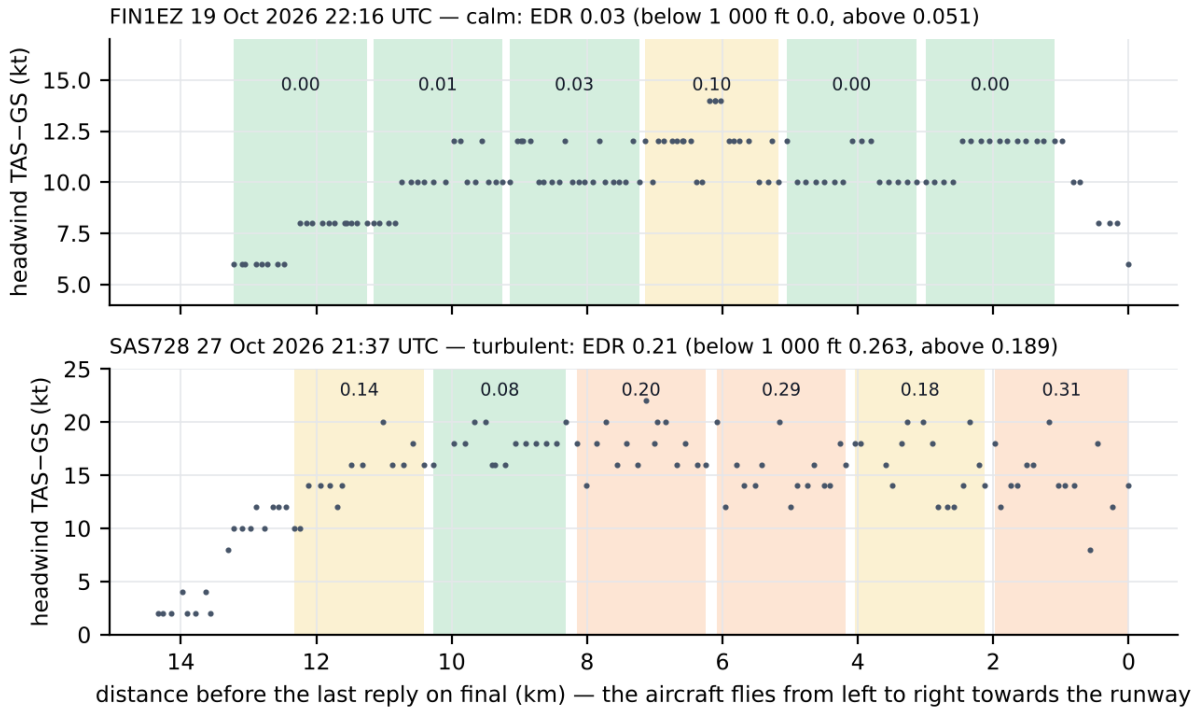


Figure 2. The headwind TAS − GS of two landings on runway 15, as received (dots), with their straight 2-km pieces coloured by their EDR. Top: a calm night (METAR wind 3 kt) — the headwind changes only in the 2-kt steps of the register; EDR 0.03. Bottom: the windy evening of 27 October 2026 (southerly 17 kt) — the headwind jumps by 4–8 kt from reply to reply; EDR 0.21, moderate below 1 000 ft (0.26), roughest piece 0.31.

6.4 The scale problem

The replies are about 140 m apart, so the shortest separation the data resolve is about 100–200 m — near the upper end of the inertial subrange close to the ground, where the large eddies are only a few hundred metres in size. Fig. 3 shows the measured structure function of all straight pieces of a calm and a windy day. It does not grow as $r^{2/3}$ beyond 200–300 m: the detrending of the 2-km pieces removes the larger scales, and in the surface layer the eddies of a few hundred metres are already the energy-containing ones. The $r^{2/3}$ law is therefore *assumed* at 100–200 m, not verified; this is the main reason the values are called a research estimate and need calibration.

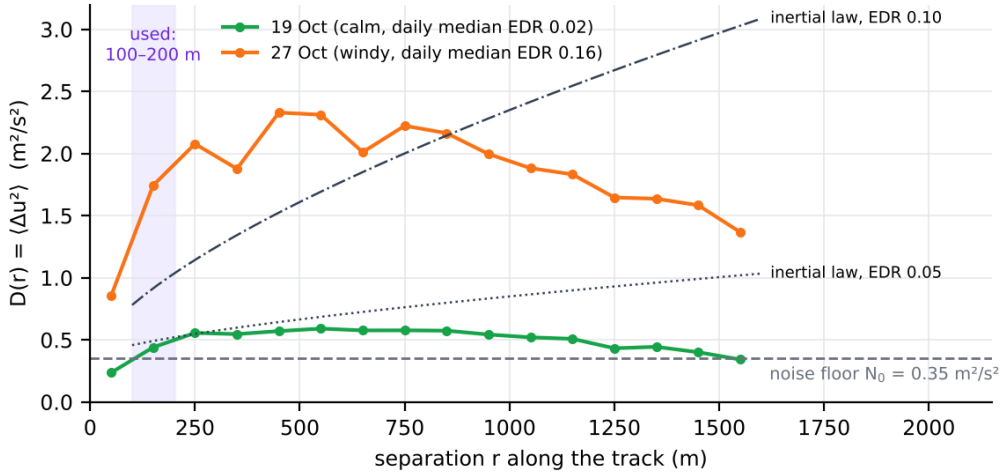


Figure 3. Structure function $D(r)$ of the headwind of all straight 2-km pieces of the landings of a calm and a windy day. The calm day stays near the noise floor N_0 of the 2-kt rounding; on the windy day the whole curve is lifted. Only the 100–200 m bin is used; the inertial-range curves for EDR 0.05 and 0.10 are shown for comparison.

6.5 Departures

A departure crosses 200–3 000 ft in about 80 s and usually turns early, so few climbs have two straight 2-km pieces below 3 000 ft. From 29 October 2026 the raw climb-out series is kept up to 5 000 ft and the departure EDR uses the climb from 200 to 5 000 ft (the value above 1 000 ft then covers 1 000–5 000 ft). Two alternatives were tested on the stored flights and rejected: the full wind vector from the heading (BDS 6,0), which lets turning pieces be used, and the turning pieces of the headwind itself; neither agreed with the straight-piece values (correlation 0.28 and 0.33).

6.6 Parameters

Quantity	Value
Signal	headwind TAS − GS of BDS 5,0, each reply (2-kt resolution)
Heights	200–3 000 ft above the threshold (landings), 200–5 000 ft (departures, from 29 Oct 2026)
Pieces	about 2 km (at least 1.5 km), track change $\leq 15°$, ≥ 8 replies, linear detrend
Separation	100–200 m (pairs), $r = 141$ m in Eq. (4)
Noise floor	$N_0 = 0.35\ \mathrm{m^2\ s^{-2}}$
Constant	$C = 2.0$ (longitudinal structure function)
Values	flight (all pieces pooled), below / above 1 000 ft, peak piece, every piece for the map
Classes	smooth $< 0.10 \leq$ light $< 0.20 \leq$ moderate $< 0.45 \leq$ severe
Times	repeated times and replies more than 3 s out of order dropped (method v2)
Code	`collector/edr.py`, method version 2; stored as `edr_json` with each landing and departure

7 Results at Helsinki-Vantaa

The figures below use all landings of 9–30 October 2026 that had a raw reply series (4 570); 3 391 (74 %) had enough straight pieces for an EDR. The rest are mainly approaches that turn onto final late, very short series, and aircraft that do not answer BDS 5,0.

Distribution. The median EDR of a landing is 0.064; the 25, 75, 90 and 99 % percentiles are 0.025, 0.106, 0.151 and 0.231 (Fig. 4). Of the landings, 28 % reach the light limit with the value of the whole approach and 2.4 % the moderate limit; with the strongest 2-km piece, closer to the ICAO peak, 51 % reach light and 14 % moderate. None reached severe. A value of 0 means that the variance was at or below the noise floor.

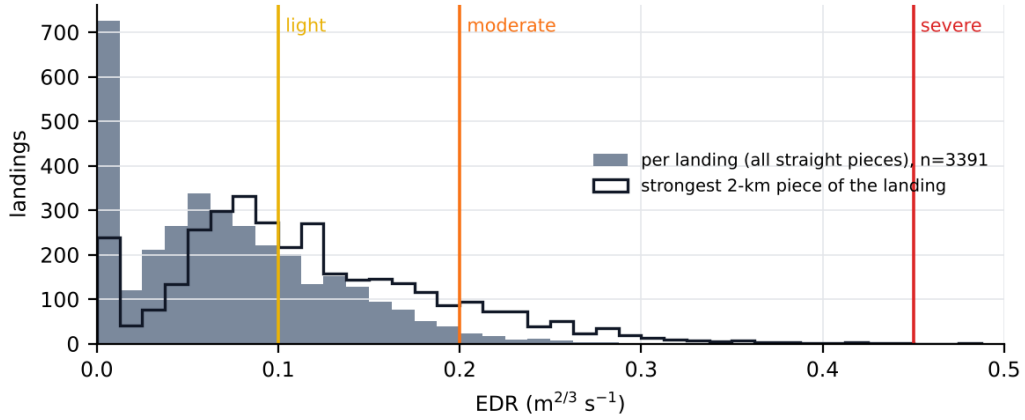


Figure 4. EDR of the 3 391 landings of 9–30 October 2026: the value of the whole approach (grey) and of its roughest 2-km piece (outline), with the ICAO limits.

Surface wind and height. EDR grows steadily with the METAR mean wind — median 0.026 at 0–4 kt, 0.042 at 5–9 kt, 0.097 at 10–14 kt and 0.150 at 15 kt and more (Spearman correlation 0.66) — and is higher below 1 000 ft (median 0.085) than above it (0.059): the mechanical turbulence of the surface layer (Fig. 5).

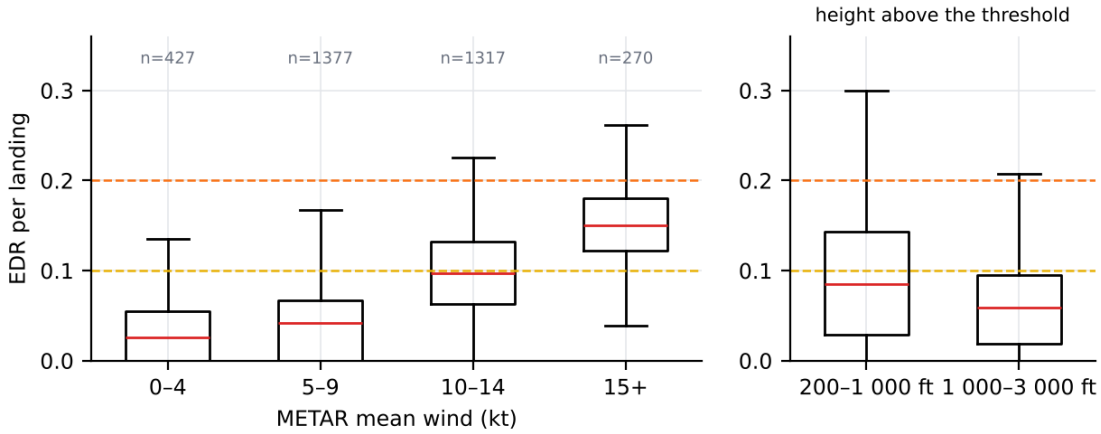


Figure 5. Left: EDR per landing against the METAR mean wind (boxes 25–75 %, whiskers 1.5 times the box, red the median). Right: below and above 1 000 ft above the threshold.

The air, not the aircraft. For each landing the ratio of its EDR to the median of the other landings within 45 minutes was formed and its median taken per aircraft class: narrow-bodies 1.03, regional jets 1.02, wide-bodies 0.97, business jets 0.98, turboprops 0.89. The Approach Conditions Index (ACI), which measures how much the aircraft itself was moved, needs class factors of 1.0–1.5 for the same comparison. EDR is nearly the same for all classes, as it should be for a property of the air; the slightly lower turboprop value may come from their lower speed (more replies per 100 m, a smaller share of the noise).

Day to day. The daily median EDR follows the daily median ACI closely (Pearson correlation 0.99 over 21 days, Fig. 6); per landing the Spearman correlation is 0.64. The windy days stand out — 16 October and 25–27 October, when southerly and south-westerly winds of 15–20 kt gave median EDRs of 0.12–0.16 — and the calm high-pressure days of 9 and 19–20 October have medians of 0.02.

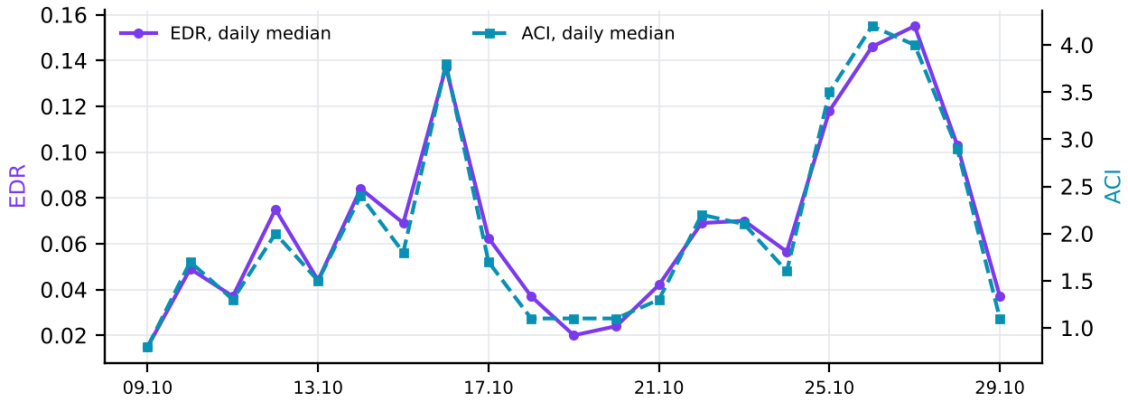


Figure 6. Daily median EDR (violet) and ACI (blue) of the landings, 9–29 October 2026 (days with at least 40 landings).

Departures. Since the climbs are followed to 5 000 ft (29 October 18:30 UTC), 77 % of the departures have an EDR (72 of 94 by 30 October 09 UTC), median 0.038 in mostly light winds. Before, only about a third had two straight pieces below 3 000 ft. The figures of this section are from method v1; the back-fill with v2 (31 October) left the landings practically unchanged (roughest piece ≥ 0.2: 470 $\rightarrow$ 469 of all stored landings) but lowered some departures that had read repeated replies (roughest piece ≥ 0.2: 14 $\rightarrow$ 8, mean ≥ 0.1: 58 $\rightarrow$ 47 of the departures of 27–31 October).

Low-level jets and inversions. In the hours with a nocturnal low-level jet the EDR below 1 000 ft was lower at the same surface wind (5–9 kt: 0.027 against 0.054; 10–14 kt: 0.086 against 0.132), and under a surface inversion it was nil — the stable, decoupled surface layer of the boundary-layer literature. The samples are still small (see the *Windshear detection* guide and the windshear study log).

8 Where EDR is shown

- **Windshear page, Wind Rose panel, EDR and ACI views**: the EDR of every landing (dot) and departure (ring) on a logarithmic axis with the ICAO limits, above the windshear strip, for 3 h to 1 day (Fig. 7).
- **Windshear page, Approach History, EDR view**: each flight's EDR as a badge in its class colour, below and above 1 000 ft and the roughest piece; a filter for light or moderate (Fig. 10).
- **Windshear page, map**: the EDR button of the map bar (Off $\rightarrow$ EDR 30 min $\rightarrow$ EDR 1 h) draws the paths of the landings of that period, each straight 2-km piece in the colour of its EDR, with a key and a summary at the lower left; the departures too when the map shows them (Departures, All 30NM or All 100NM, from 1 November 2026).
- **One flight's headwind profile** (from 31 October 2026): a click on a landing or departure in the Approach History opens its headwind from the ground up against the flights before it, with the EDR of each piece at the right edge; on the phone the same chart heads an opened History card.
- **Phone view `/app`**: the EDR strip of the EDR · WS · ACI · ST view (named ACI · WS · ST until 1 November 2026; also its EDR + WS chart) and of the chart under the ILS view, an EDR chip on each History card — and on the flight cards of the EDR · WS · ACI · ST view — and Turbulence on the map: every straight piece of the paths of the last 30 min or 1 h coloured by its EDR (Fig. 8).
- **Boundary layer**: the EDR hover of the Approach History and the opened History cards of the phone give the low-level jet and the surface inversion of the hour of each flight, for comparison with its EDR; the BL view shows them hour by hour.



Figure 7. Windshear page, EDR view, 6 hours of 28 October 2026 replayed in a test: most landings light, the departures (rings) a little lower; no windshear alerts below.

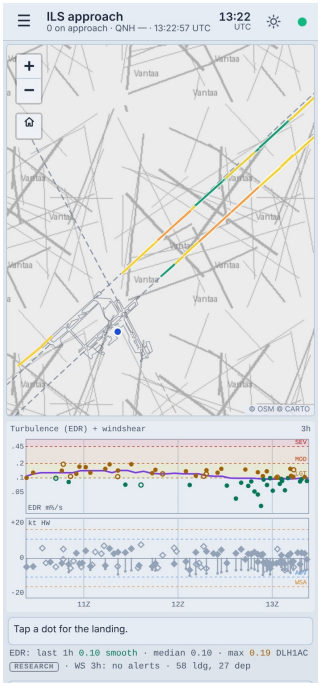


Figure 8. Phone, Turbulence on the map, 28 October 2026 12:22–13:22 UTC replayed: 32 landings, 2 departures, peak 0.26.

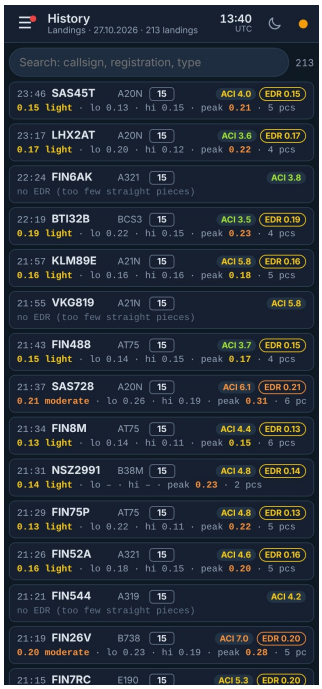


Figure 9. Phone, History with Show: Turbulence, the landings of 27 October 2026.

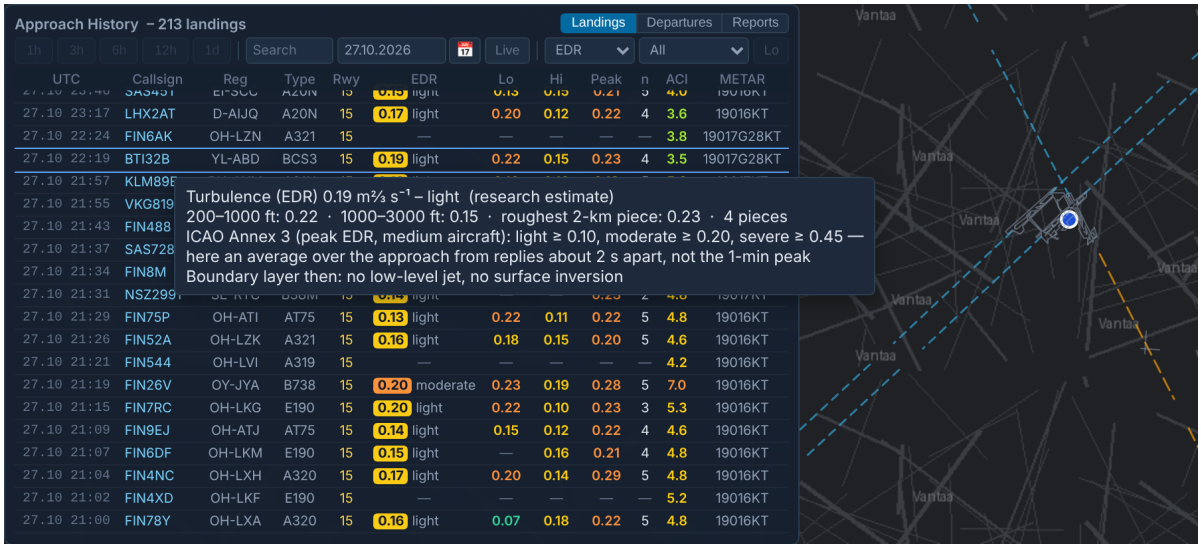


Figure 10. Approach History, EDR view, 27 October 2026 (southerly 16–17 kt, gusts to 28 kt): most landings light, SAS728 (Fig. 2) and FIN26V moderate below 1 000 ft.

9 Limitations and calibration

- **Scale.** Only 100–200 m is resolved; the inertial-range law is assumed there (Section 6). Values from other systems that use shorter scales (aircraft accelerometers at 8 Hz or more, lidar gates of tens of metres) need not agree in absolute level.
- **Mean, not peak.** A flight value is an average over many pieces; the ICAO limits refer to one-minute peaks. The roughest 2-km piece is closer to a peak but is still an average over about 15 seconds.
- **One component.** Only the along-track (longitudinal) component is measured; vertical gusts, which shake the aircraft most, are not seen.
- **Quantisation.** The 2-kt rounding sets a floor of about 0.03–0.04 below which values are not meaningful; 0 (shown as <0.01) means "at or below the floor".
- **Coverage.** About a quarter of the landings and of the departures have no value (late turns onto final, short series, no

BDS 5,0). In addition, an arrival whose GNSS is still jammed at touchdown gets no approach record at all, and so no EDR: 5.6 % of the arrivals on 7 October – 1 November 2026, but 11–14 % at 21–02 UTC, mostly arrivals from the jammed directions — the night values rest on fewer landings. The EDR of the recorded landings is not affected (it uses the headwind replies, not the positions). If the jamming ends, the gap closes by itself.

- **Calibration.** The method has not yet been compared with in-situ EDR from aircraft or with pilot reports. From 29 October 2026 MODE-S Wind stores the special air reports (ARS) and the Finnish WXREP pilot reports near the airport for this purpose; until then the classes "light" and "moderate" indicate the ICAO limits applied to this estimate, not a verified match. By 1 November 2026, in calm high-pressure weather, only two ARS had been stored, both of moderate icing (one near the airport at 3 000–4 000 ft) and none of turbulence; the comparison needs the windy months ahead.

10 Version history

Date	Document / method	Change
29 Oct 2026	method v1	First version: straight 2-km pieces, 100–200 m structure function, $N_0 = 0.35$, Eq. (4); back-filled for all stored landings (from 9 October) and departures. Departures 200–5 000 ft.
30 Oct 2026	document 1.0	First edition, results of 9–30 October 2026.
31 Oct 2026	method v2	Repeated reply times and replies more than 3 s out of order dropped before the pieces; all stored landings and departures back-filled (Section 6).
1 Nov 2026	document 1.1	Method v2; the Windshear map, the one-flight headwind profile and the phone's EDR · WS · ACI · ST view among the displays; the first pilot reports.
1 Nov 2026	document 1.2	Coverage: the landings left without a record by GNSS jamming.

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